Order of Operations

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KEY WORDS order of operations *(raupapa mahi)* · GEMA · grouping · brackets *(taiapa)* · exponent *(taupū)*

A Multiplication and division first ●○○

⚠ **Watch out:** Do not just work from left to right. Multiply and divide before you add and subtract: $3 + 4 \times 5 = 3 + 20 = 23$.

1. Work out.

a) $3 + 4 \times 5$ = _____ **b)** $20 - 12 \div 4$ = _____ **c)** $6 \times 2 + 7$ = _____ **d)** $18 \div 3 - 2$ = _____

e) $10 - 2 \times 3$ = _____ **f)** $4 + 16 \div 8$ = _____ **g)** $5 \times 3 + 2 \times 4$ = _____ **h)** $30 - 5 \times 5$ = _____

i) $7 + 21 \div 7$ = _____ **j)** $9 \times 2 - 12 \div 4$ = _____ **k)** $40 \div 8 + 6 \times 2$ = _____ **l)** $15 - 3 \times 4 + 6$ = _____

B Brackets ●●○

💡 **Remember:** GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

2. Work out.

a) $(3 + 4) \times 5$ = _____ **b)** $18 \div (2 + 4)$ = _____ **c)** $(10 - 4) \times 3$ = _____

d) $2 \times (9 - 5)$ = _____ **e)** $(8 + 12) \div 5$ = _____ **f)** $30 - (7 + 8)$ = _____

g) $(6 - 2) \times (3 + 4)$ = _____ **h)** $5 + 2 \times (8 - 3)$ = _____ **i)** $(15 - 3) \div (1 + 3)$ = _____

j) $4 \times (2 + 5) - 8$ = _____ **k)** $50 \div (12 - 2) + 7$ = _____ **l)** $(20 - 2 \times 5) \div 2$ = _____

C Exponents ●●○

⚠ **Watch out:** Work out an exponent before you multiply: $2 \times 3^2 = 2 \times 9 = 18$, not 6^2.

3. Work out. Show your first step.

a) 2×3^2 **b)** $4^2 - 3 \times 5$ **c)** $5 + 2^3$

d) $(6 + 2)^2 \div 4$ **e)** $50 - 5^2 \times 2$ **f)** $3^2 + 4^2$

g) $(3 + 4)^2$ **h)** $10^2 \div (2 + 3)$ **i)** $2^3 \times (7 - 4)$

j) 4×5^2 **k)** $2^4 - 2^3$ **l)** $(10 - 7)^3 + 1$

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D With negative numbers ●●●

4. Work out. Show your first step.

a) $-3 + 4 \times 2$ **b)** $5 - 2 \times 6$ **c)** $(-4 + 9) \times 3$

d) $-10 \div 2 + 7$ **e)** $6 + (-2) \times 5$ **f)** $(2 - 8) \times 3$

g) $-4 \times 5 + 30$ **h)** $12 - (3 - 7)$ **i)** $-20 \div 4 - 6$

j) $(-3)^2 + 1$ **k)** $2 \times (-5) - 4$ **l)** $-8 + 12 \div (-4)$

E Mixed practice ●●●

5. Work out.

a) $2 + 3 \times 4 - 5$ = _____ **b)** $(2 + 3) \times (4 - 1)$ = _____ **c)** $6^2 \div 4 + 1$ = _____

d) $100 - 5 \times (2 + 6)$ = _____ **e)** $3 \times 2^2 + 10$ = _____ **f)** $48 \div (2 \times 4)$ = _____

g) $(7 - 3)^2 + 5$ = _____ **h)** $8 + 12 \div 4 \times 3$ = _____ **i)** $5 \times (10 - 2 \times 3)$ = _____

j) $2^3 + 3^2 - 1$ = _____ **k)** $60 \div 5 \div 2$ = _____ **l)** $20 - 8 + 5$ = _____

F Problems ●●●

6. Write one calculation for each problem, then work it out.

a) Mia buys 3 pies at $5 each and one drink for $4. How much does she spend?

b) Tane has $50. He buys 6 tickets at $7 each. How much money does he have left?

c) One class has 24 students and another has 18. All the students are put into 6 equal teams. How many are in each team?

d) A shop sells 4 boxes of 12 eggs and 3 boxes of 8 eggs. How many eggs is that altogether?

e) A square room has sides of 5 m. A rug covers 9 m² of the floor. A second room is exactly the same. How much floor is not covered by the rugs in the two rooms altogether?

f) A prize of $60 is shared equally between 4 friends. Each friend then spends $3 on a drink. How much does each friend have left?

★ Extension

7. Use the numbers 2, 3 and 5 once each, with any operations and brackets, to make 25.

8. Now use the same three numbers to make 11.

9. Put one pair of brackets into each calculation to make it true.

a) $3 + 4 \times 5 = 35$ **b)** $20 - 12 \div 4 + 2 = 18$ **c)** $2 \times 5 - 1^2 = 32$

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1-1 PRIVATE TUTORS

NUMBER

Order of Operations

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS order of operations *(raupapa mahi)* · GEMA · grouping · brackets *(taiapa)* · exponent *(taupū)*

A Multiplication and division first ●○○

⚠ **Watch out:** Do not just work from left to right. Multiply and divide before you add and subtract: $3 + 4 \times 5 = 3 + 20 = 23$.

1. Work out.

a) $2 + 3 \times 6$ = _____ **b)** $30 - 18 \div 6$ = _____ **c)** $5 \times 3 + 8$ = _____ **d)** $24 \div 4 - 3$ = _____

e) $12 - 2 \times 4$ = _____ **f)** $5 + 20 \div 10$ = _____ **g)** $4 \times 3 + 3 \times 5$ = _____ **h)** $40 - 6 \times 6$ = _____

i) $8 + 24 \div 8$ = _____ **j)** $8 \times 2 - 15 \div 3$ = _____ **k)** $36 \div 9 + 7 \times 2$ = _____ **l)** $20 - 4 \times 4 + 5$ = _____

B Brackets ●●○

💡 **Remember:** GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

2. Work out.

a) $(2 + 3) \times 6$ = _____ **b)** $24 \div (3 + 5)$ = _____ **c)** $(12 - 5) \times 4$ = _____

d) $3 \times (8 - 2)$ = _____ **e)** $(9 + 15) \div 6$ = _____ **f)** $40 - (9 + 6)$ = _____

g) $(8 - 5) \times (3 + 4)$ = _____ **h)** $4 + 3 \times (9 - 5)$ = _____ **i)** $(20 - 2) \div (4 + 2)$ = _____

j) $5 \times (1 + 3) - 9$ = _____ **k)** $60 \div (15 - 3) + 8$ = _____ **l)** $(30 - 4 \times 3) \div 3$ = _____

C Exponents ●●○

⚠ **Watch out:** Work out an exponent before you multiply: $2 \times 3^2 = 2 \times 9 = 18$, not 6^2.

3. Work out. Show your first step.

a) 3×2^3 **b)** $5^2 - 4 \times 3$ **c)** $6 + 3^2$

d) $(5 + 1)^2 \div 9$ **e)** $36 - 3^2 \times 4$ **f)** $6^2 + 8^2$

g) $(2 + 6)^2$ **h)** $10^2 \div (1 + 4)$ **i)** $3^2 \times (9 - 5)$

j) 5×4^2 **k)** $3^3 - 3^2$ **l)** $(9 - 7)^3 + 2$

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D With negative numbers ●●●

4. Work out. Show your first step.

a) $-5 + 3 \times 2$ **b)** $4 - 3 \times 5$ **c)** $(-3 + 9) \times 2$

d) $-12 \div 3 + 9$ **e)** $8 + (-3) \times 4$ **f)** $(3 - 8) \times 4$

g) $-6 \times 4 + 40$ **h)** $15 - (2 - 9)$ **i)** $-30 \div 5 - 4$

j) $(-4)^2 + 2$ **k)** $3 \times (-4) - 5$ **l)** $-9 + 15 \div (-3)$

E Mixed practice ●●●

5. Work out.

a) $3 + 2 \times 5 - 4$ = _____ **b)** $(3 + 4) \times (5 - 2)$ = _____ **c)** $8^2 \div 4 + 3$ = _____

d) $100 - 4 \times (3 + 7)$ = _____ **e)** $2 \times 3^2 + 20$ = _____ **f)** $54 \div (3 \times 3)$ = _____

g) $(9 - 4)^2 + 6$ = _____ **h)** $6 + 16 \div 4 \times 2$ = _____ **i)** $4 \times (12 - 3 \times 3)$ = _____

j) $3^2 + 2^4 - 1$ = _____ **k)** $80 \div 4 \div 2$ = _____ **l)** $30 - 9 + 4$ = _____

F Problems ●●●

6. Write one calculation for each problem, then work it out.

a) Mia buys 4 pies at $6 each and one drink for $3. How much does she spend?

b) Tane has $100. He buys 8 tickets at $9 each. How much money does he have left?

c) One class has 35 students and another has 21. All the students are put into 7 equal teams. How many are in each team?

d) A shop sells 5 boxes of 10 eggs and 2 boxes of 15 eggs. How many eggs is that altogether?

e) A square room has sides of 6 m. A rug covers 7 m² of the floor. A second room is exactly the same. How much floor is not covered by the rugs in the two rooms altogether?

f) A prize of $80 is shared equally between 5 friends. Each friend then spends $2 on a drink. How much does each friend have left?

★ Extension

7. Use the numbers 2, 4 and 6 once each, with any operations and brackets, to make 36.

8. Now use the same three numbers to make 14.

9. Put one pair of brackets into each calculation to make it true.

a) $2 + 6 \times 4 = 32$ **b)** $30 - 18 \div 6 + 3 = 28$ **c)** $3 \times 4 - 2^2 = 12$

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